

Penrose and TNA: Consciousness, Non-Closure, and the Limits of Local Derivability

A Comparative Structural Study

Abstract

This paper presents a comparative analysis between the philosophical implications of the work of Roger Penrose and the framework proposed by the Theory of Axiomatic Necessity (TNA).

Although developed independently and from different domains, both frameworks converge on a critical ontological intuition: reality may not be fully derivable from locally closed computational or physical dynamics alone.

Penrose argues that consciousness cannot be reduced to algorithmic computation or classical neuronal processing. TNA generalizes this principle by proposing that sufficiently complex realizable systems cannot fully generate their own conditions of closure and legitimacy from internal local dynamics (N_0) alone, requiring external selection structures (N_1).

This paper compares Penrose's views on consciousness, computation, Gödel incompleteness, quantum collapse, and cosmology with TNA concepts such as Failure of Local Closure, non-derivable realizability, structural openness, and continuous but non-smooth support structures.

The study suggests that both approaches represent parallel attempts to overcome the limitations of strict reductionism, although TNA extends the principle of non-closure beyond consciousness into civilization, metaphysics, legitimacy, and cosmological realizability itself.

1. Introduction

One of the defining assumptions of modern reductionism is that sufficiently complex local dynamics eventually become self-explanatory.

Under this paradigm:

- consciousness becomes computation,

- meaning becomes emergent pattern recognition,
- civilization becomes optimization,
- and cosmology becomes self-contained physical propagation.

Both Penrose and TNA challenge this assumption.

While Penrose approaches the problem through mathematical physics and consciousness studies, TNA approaches it through realizability theory and structural non-closure.

Yet both frameworks converge on a common claim:

local dynamics alone may be insufficient to fully explain reality.

2. Penrose's Core Position

Penrose's critique of computational reductionism rests on several interconnected ideas:

2.1 Gödelian Non-Computability

Drawing from Kurt Gödel, Penrose argues that human mathematical understanding exceeds formal algorithmic systems.

Humans appear capable of recognizing truths that cannot be fully derived inside closed symbolic systems.

Therefore:

- mind cannot be purely computational,
 - and consciousness may involve non-computable processes.
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2.2 Consciousness and Quantum Reduction

Together with Stuart Hameroff, Penrose proposed the Orch-OR model:

- consciousness emerges from orchestrated quantum collapses,
- potentially involving gravity-induced objective reduction,
- occurring inside neuronal microtubules.

This suggests:

- subjective experience is linked to fundamental physics,
 - not merely large-scale computation.
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2.3 Conformal Cyclic Cosmology (CCC)

Penrose's cosmological model proposes:

- eternal cycles (“aeons”),
- conformal rescaling at heat death,
- black hole evaporation,
- and renewed Big Bang conditions.

CCC attempts to eliminate the need for an absolute beginning while preserving mathematical continuity.

Yet the theory introduces a deep question:

what guarantees the realizability of the transition itself?

3. The Core Structure of TNA

TNA distinguishes between two structural domains:

3.1 N_0 — Local Dynamics

N_0 includes:

- physical propagation,
- biological evolution,
- optimization,
- neural computation,
- causal continuity,
- operational state transitions.

N_0 is locally derivable and dynamically smooth.

3.2 N_1 — External Realizability Support

N_1 includes:

- legitimacy,
- realization constraints,
- external selection,
- closure conditions,
- transcendence,
- structural intelligibility.

N_1 is not another physical force.

It is the condition that stabilizes realizability when local multiplicity becomes insufficiently self-selecting.

3.3 Failure of Local Closure

Failure of Local Closure occurs when:

- multiple admissible trajectories exist ($\Delta > 1$),
- but local dynamics alone cannot uniquely stabilize realization.

TNA proposes that this condition appears in:

- consciousness,
 - morality,
 - civilization,
 - cosmology,
 - identity,
 - and existential meaning.
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4. Comparative Analysis: Penrose vs TNA

4.1 Consciousness

Penrose

Consciousness cannot be reduced to classical computation.

TNA

Consciousness cannot fully emerge from internally closed No dynamics.

Convergence

Both reject strong computationalism.

Both imply:

subjective experience exceeds local algorithmic closure.

4.2 Gödel and Non-Closure

Penrose

Gödel incompleteness demonstrates limits of formal systems.

TNA

Failure of Local Closure generalizes incompleteness into realizability itself.

Difference

Penrose applies incompleteness primarily to mind and mathematics.

TNA extends it toward:

- civilization,
 - ontology,
 - legitimacy,
 - and cosmological structure.
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5. Quantum Collapse and N_1

Penrose treats objective reduction as physically fundamental.

TNA can reinterpret this structurally:

- quantum collapse represents a selection event,
- where multiplicity becomes constrained into realizability.

Under this reading:

- collapse resembles an N_1 -like selection operator,
- though not necessarily theological or supernatural.

This does not imply “magic.”

Rather:

- local propagation alone may be insufficient to explain realization selection.
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6. CCC and Structural Reset

Penrose’s CCC attempts to explain cosmic renewal entirely from internal cosmological dynamics.

However, TNA identifies a structural tension:

the transition between aeons behaves functionally like an external realizability condition.

The conformal reset:

- removes entropy,
- removes scale,

- restores realizability,
- and permits renewed cosmic structure.

From a TNA perspective:

CCC may implicitly require an N_i -type structural operator even while attempting complete cosmological self-closure.

7. The Fundamental Difference

Despite similarities, Penrose and TNA diverge fundamentally.

Penrose

Still seeks:

- a deeper physics,
- a mathematically unified explanation,
- and potentially an internal cosmological completion.

TNA

Argues that:

- complete self-closure may be structurally impossible in principle.

TNA therefore treats:

- transcendence,
- external legitimacy,
- and non-derivable support

not as temporary gaps in science,
but as permanent structural features of realizability.

8. Consciousness as Structural Openness

Under TNA, consciousness may be interpreted not merely as computation plus complexity, but as:

openness toward non-local realizability support.

This interpretation explains why:

- humans pursue transcendence,
- sacrifice utility,
- search for meaning,
- experience existential rupture,
- and question the legitimacy of reality itself.

AI systems, by contrast, remain operationally closed inside predefined state spaces.

They may generate multiplicity,
but not structural openness toward N_1 .

9. Cosmological Implications

Both Penrose and TNA reject simplistic materialism.

Both imply that:

- reality contains deeper structural layers,
- and that observable physics may not fully explain existence.

Yet TNA pushes further:

If even cosmology requires non-local realizability conditions,
then transcendence is not merely religious speculation,
but structurally compatible with the ontology of open systems.

10. Final Reflection

Penrose represents one of the most important modern attempts to challenge reductionism from inside mathematics and physics.

TNA generalizes that challenge.

Where Penrose identifies:

- non-computability,
- quantum indeterminacy,
- and cosmological incompleteness,

TNA identifies a broader principle:

no sufficiently complex realizable system can fully generate its own closure conditions from local dynamics alone.

In this sense, Penrose and TNA may be understood as parallel frameworks converging toward the same frontier:

the limits of self-contained reality.

Conclusion

This paper explored the structural convergence between Penrose's philosophy of consciousness and cosmology and the Theory of Axiomatic Necessity.

Both frameworks reject strict reductionism and challenge the assumption that local computation or physical propagation can fully explain consciousness, realization, or existence.

Penrose approaches the problem through:

- Gödel,
- quantum collapse,
- and cosmological cyclicity.

TNA approaches it through:

- Failure of Local Closure,

- realizability theory,
- and the distinction between N_0 and N_1 .

The comparison suggests that both theories point toward a deeper ontological insight:

reality may remain fundamentally open, non-self-sufficient, and structurally incapable of total closure.

Under this interpretation:

- consciousness,
- transcendence,
- meaning,
- and cosmological realization

are not anomalies to eliminate,
but signatures of a universe whose intelligibility exceeds purely local derivation.

Structural correspondence Penrose $\leftrightarrow$ TNA:

- *Gödel incompleteness $\leftrightarrow$ Failure of Local Closure*
- *Objective reduction (Orch-OR) $\leftrightarrow$ N_1 -selection event*
- *CCC conformal reset $\leftrightarrow$ N_1 -cosmologic reset operator*
- *Non-computability of mind $\leftrightarrow$ Non-derivability of N_1 from N_0*

References

- Penrose, R. (1989). *The Emperor's New Mind: Concerning Computers, Minds and the Laws of Physics*. Oxford University Press.
- Penrose, R. (1994). *Shadows of the Mind: A Search for the Missing Science of Consciousness*. Oxford University Press.
- Penrose, R. (1997). *The Large, the Small and the Human Mind*. Cambridge University Press.
- Hameroff, S. & Penrose, R. (2014). "Consciousness in the universe: A review of the 'Orch OR' theory." *Physics of Life Reviews*, 11(1), 39-78.
- Gödel, K. (1931). "Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I." *Monatshefte für Mathematik und Physik*, 38, 173-198.
- Putnam, H. (1995). "Review of *Shadows of the Mind*." *Bulletin of the American Mathematical Society*, 32(3), 370-373. (Para la crítica al argumento gödeliano)

- Bresciano, Claudio (2025). Zenodo. DOI 10.5281/zenodo.19192655. TNA (Failure of Local Closure).
- Bresciano, Claudio (2026). Zenodo. DOI 10.5281/zenodo.20224880. Continuous but Non-Differentiable Support.